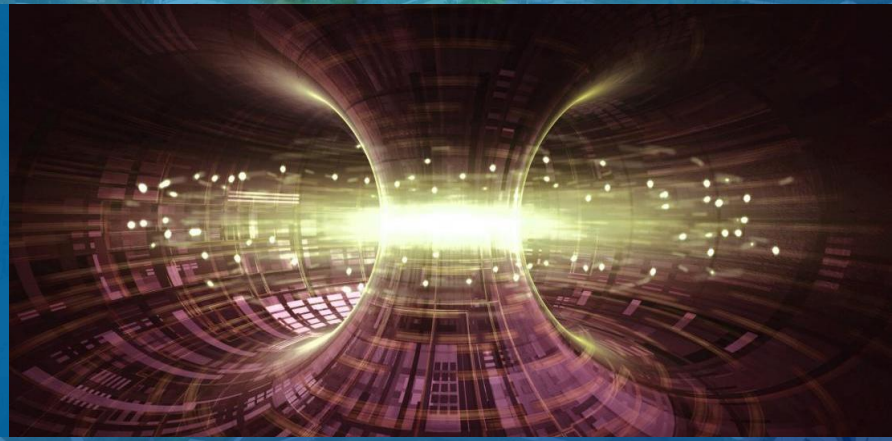
ANL Learning Off The Lawn

July 29, 2026

Multi-Agent AI System for PETSc Simulation of Plasma in a Tokamak (for Nuclear Fusion Energy)



Presenter
Sarthak Sharma[1,2]
Intern and PhD student

Advisors
Dr Junchao Zhang[1], Dr Lois Curfman McInnes[1], Prof Matthew Knepley[2].

[1]Division of Mathematics and Computer Science, Argonne National Laboratory,
[2]Computational and Data Sciences, State University of New York at Buffalo.

U.S. DEPARTMENT of ENERGY

Argonne National Laboratory is a U.S. Department of Energy laboratory managed by UChicago Argonne, LLC.

University at Buffalo The State University of New York

PETSc TAO

Argonne NATIONAL LABORATORY

Contents

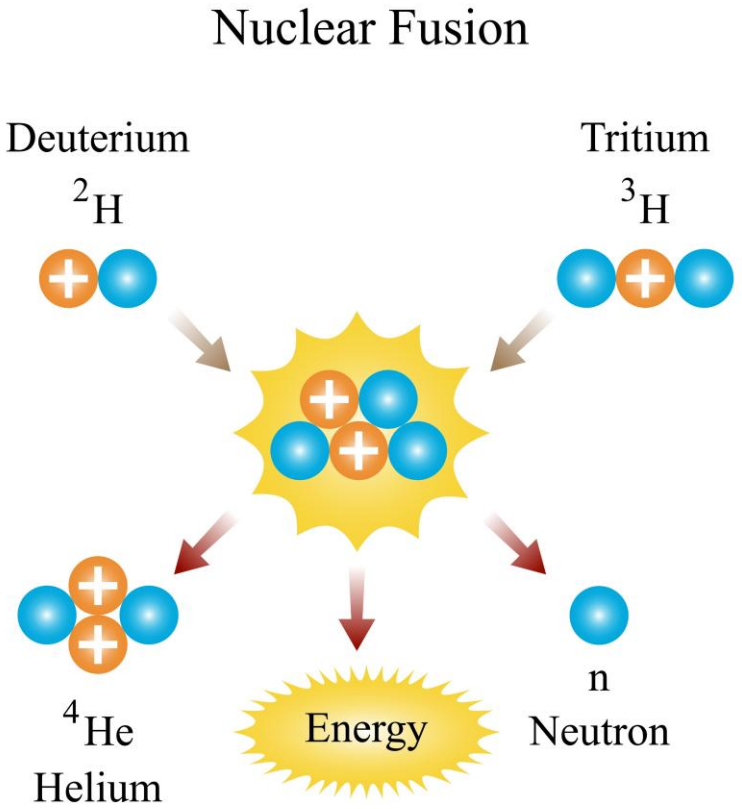
Why nuclear fusion energy needs trustworthy simulation

Nuclear fusion could be a near-limitless clean energy source, if we can solve some challenges.

- Combine light nuclei to create heavy nuclei (e.g., hydrogen isotopes converted to helium).
- Small amount of mass converted to large amount of energy:

$$E = mc^2$$

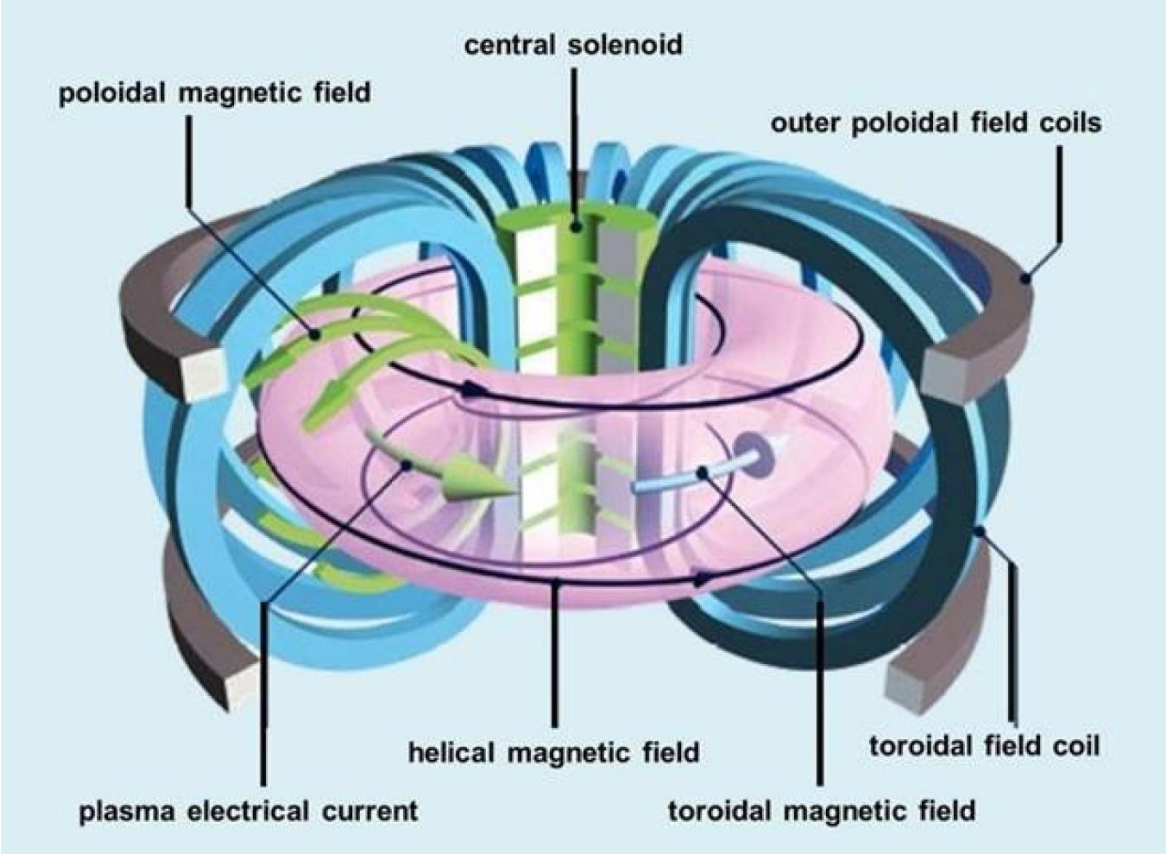
- Source of energy in stars.



Deuterium–tritium fusion

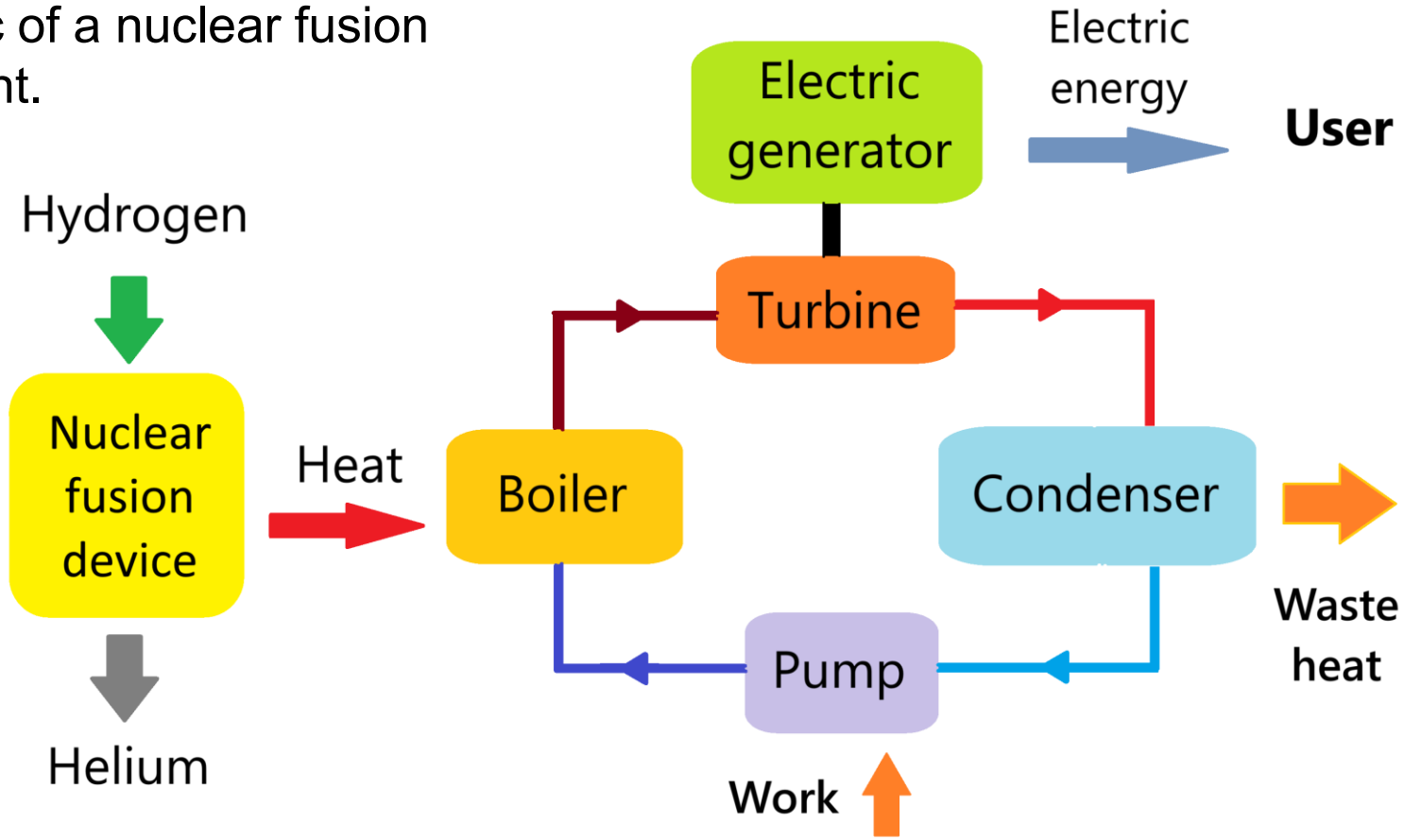
Why nuclear fusion energy needs trustworthy simulation (continued)

- Tokamak confines plasma (temperature around 150 million °C) in a magnetic “cage” long enough to fuse.
- Accurate and scalable simulations needed before expensive and time-consuming physical experiments.
- Simulations must be trustworthy. Plausible-looking wrong answer is worse than no answer.



Tokamak magnetic confinement (U.S. DOE)

Schematic of a nuclear fusion power plant.



The modeling challenge

Turning fusion physics into correct and fast HPC code needs scarce and implicit expertise.

- **Numerical methods:** Choose grids, discretizations, and solvers that actually converge.
- **Library:** Express it in PETSc (Portable, Extensible Toolkit for Scientific Computation), a library for high-performance computing.
- **Parallelism:** Run on many CPU cores / GPUs.
- **Verification:** Prove that the computed answer is right, not merely plausible.
- Question: Can a multi-agent AI system automate this whole path, and can we trust it?

The problem: Grad-Shafranov equilibrium

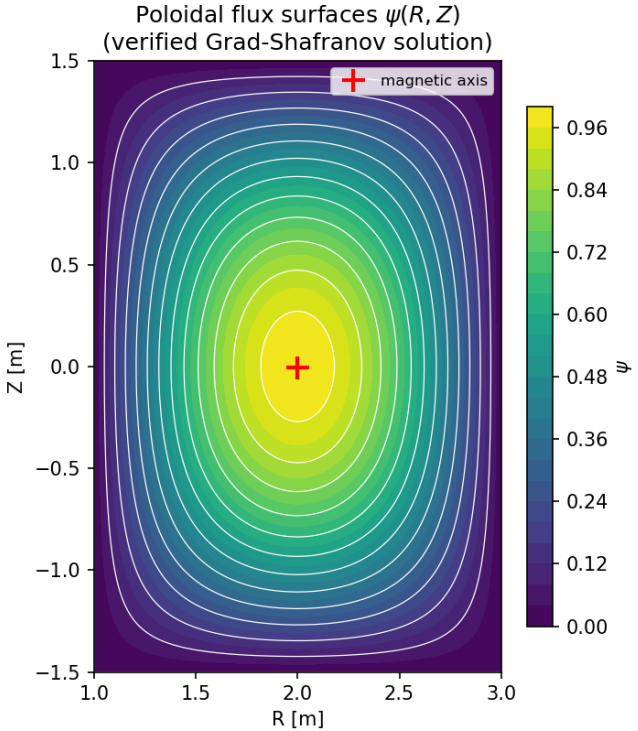
Axisymmetric ideal-MHD force balance for the poloidal flux $\psi(r, Z)$.

- Equilibrium magnetohydrodynamics for 2-dimensional plasma in a tokamak:

$$\frac{\partial^2 \psi}{\partial r^2} - \frac{1}{r}\frac{\partial \psi}{\partial r} + \frac{\partial^2 \psi}{\partial z^2} = -\mu_0 r^2 \frac{dp}{d\psi} - \frac{1}{2}\frac{dF^2}{d\psi}$$

where ψ is magnetic flux, p is plasma pressure, μ_0 is magnetic permeability, $F = rB_\theta$, B is magnetic field, and (r, θ, z) are cylindrical coordinates.

- Elliptic, non-linear, time-independent PDE. Its solution gives the flux surfaces, magnetic axis, and safety factor q.
- A natural problem for PETSc SNES (Scalable Nonlinear Equations Solvers).
- Verifiable: a manufactured exact solution gives a rigorous convergence test.

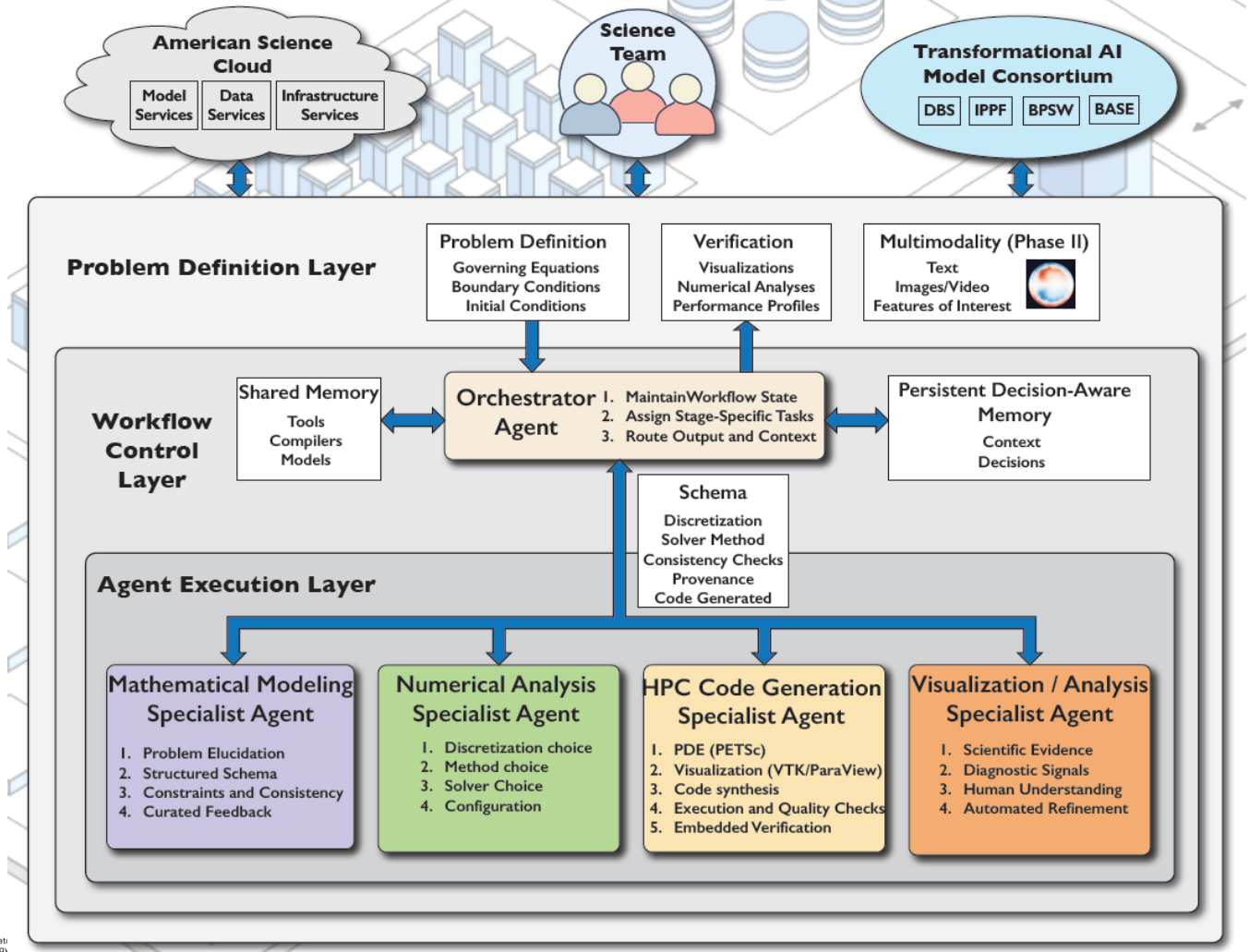


Flux surfaces of the verified solution

The AI system: 3-layer multi-agent AI pipeline

PETSc Model-Context-Protocol (MCP) agents, coordinated end to end.

- Problem definition (**Mathematical Modeling agent**): strong / weak form, problem name, time-dependence.
- Agent execution (**Numerical Analysis agent**): grid, discretization, solver (SNES).
- Agent execution (**HPC Code Generation agent**): writes, compiles and runs PETSc C programs.
- Execution substrate (**Compile and Run agent**): make / run on the real machine.
- All agents run against Argonne's Argo gateway (Claude Opus 4.8).
- A project-owned driver records every artifact with provenance (reproducible and resumable).


American Science Cloud
Model Services
Data Services
Infrastructure Services
Science Team
Transformational AI Model Consortium
DBS
IPPF
BPSW
BASE
Problem Definition Layer
Problem Definition
Governing Equations
Boundary Conditions
Initial Conditions
Verification
Visualizations
Numerical Analyses
Performance Profiles
Multimodality (Phase II)
Text
Images/Video
Features of Interest
Workflow Control Layer
Shared Memory
Tools
Compilers
Models
Orchestrator Agent
1. MaintainWorkflow State
2. Assign Stage-Specific Tasks
3. Route Output and Context
Persistent Decision-Aware Memory
Context
Decisions
Schema
Discretization
Solver Method
Consistency Checks
Provenance
Code Generated
Agent Execution Layer
Mathematical Modeling Specialist Agent
1. Problem Elucidation
2. Structured Schema
3. Constraints and Consistency
4. Curated Feedback
Numerical Analysis Specialist Agent
1. Discretization choice
2. Method choice
3. Solver Choice
4. Configuration
HPC Code Generation Specialist Agent
1. PDE (PETSc)
2. Visualization (VTK/ParaView)
3. Code synthesis
4. Execution and Quality Checks
5. Embedded Verification
Visualization / Analysis Specialist Agent
1. Scientific Evidence
2. Diagnostic Signals
3. Human Understanding
4. Automated Refinement

The AI pipeline in action

From one English prompt to a compiled, running solver, with no human edits.

- **Prompt:** "*the Grad-Shafranov equilibrium for the plasma in a tokamak*."
- **Modeling AI agent:** identified 'Grad-Shafranov equation'; time-dependent = False.
- **Numerical Analysis AI agent:** non-linear solve with SNES on a structured grid.
- **Code Generation AI agent:** 267-line PETSc program (DMDA + SNES + true Jacobian).
- Compiled and ran on 1 and 4 MPI ranks. No human edits.

AI-agent-generated PETSc solver (excerpt)

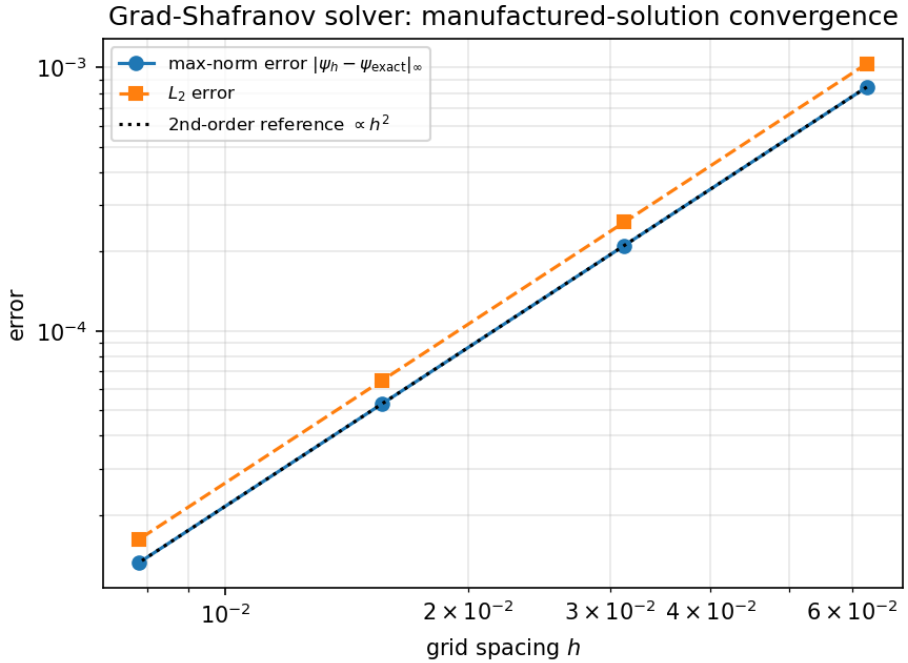
A true-Jacobian finite-difference SNES residual, written by the Code Generation AI agent.

```
static PetscErrorCode FormFunction(SNES snes, Vec X, Vec F, void *ctx){
  /* Delta*_h psi = (psi_E-2psi_C+psi_W)/hR^2
                  - (1/R)(psi_E-psi_W)/(2hR)
                  + (psi_N-2psi_C+psi_S)/hZ^2   */
  f[j][i] = dRR - dR1/R + dZZ - ForcingF(user,R,Z);   /* residual */
}
SNESSetFunction(snes, r, FormFunction, &user);
SNESSetJacobian(snes, J, J, FormJacobian, &user);   /* true Jacobian */
SNESSolve(snes, NULL, x);
```

Verification

Method of manufactured solutions: a measured order of accuracy, not just plausible output.

- Known exact ψ: check the numerical error as the grid refines.
- Max-norm error = 1.32×10^{-5} on the finest 257^2 grid.
- Observed order of accuracy $p = 2.00, 2.00, 2.00$ (expected 2 for central differences).
- SNES converged: CONVERGED_FNORM_RELATIVE.



Manufactured-solution convergence

Decision-gate metrics

Correctness, human effort, and efficiency: the numbers behind the demo.

- **Correctness**:
  - — model identified,
  - — C programs compiled and ran,
  - — second-order convergence.
- **Human effort**: 0 lines of solver code hand-written. It was all AI-generated.
- **Efficiency**: total wall-clock ≈ 450.8 seconds; code-gen used 5 tool calls.
- Around 23 completions of LLM (large language model) across the whole pipeline.

Fully autonomous: the built-in AI orchestrator

The orchestrator AI agent drives all four specialist AI agents itself (no need for a human to micro-manage).

- Given only the prompt, an inner Claude AI agent sequenced all four MCP servers on its own.
- It independently chose a Solov'ev Grad–Shafranov model and a DMPLEX + PetscFE finite-element solve.
- The AI-generated code compiled cleanly and ran: 225 DOFs, L2 norm $\psi = 2.04$, return code 0.
- It surfaced a real upstream bug: a hard-coded iteration cap flagged a converged run as failure. We fixed it and the re-run finished: "`I have completed the orchestration.`"
- The two orchestration paths independently chose two different valid PETSc discretizations (DMDA in testing specialist agents, vs DMPLEX in built-in orchestrator) for the same problem.

What we built and contributed

Reusable tooling around the agents, plus fixes contributed back upstream.

- A project-owned orchestration driver with full artifact provenance (resumable).
- Verification + post-processing (convergence, flux surfaces) and a metrics harness.
- Four fixes contributed back to the open PETSc multi-agent system:
  - CWD-independent server resolution; portable `stdio` interpreter.
  - Graceful operation without the documentation / RAG services.
  - Code-generation and orchestrator loops that reliably capture results.

Takeaways and next steps

A verification-driven multi-agent path from prompt to trustworthy HPC code

- A multi-agent AI system generated a correct, verified PETSc fusion-MHD solver from a prompt.
- Both the Python driver and the fully-autonomous orchestrator succeeded end to end.
- Verification-driven: convergence + exact-solution checks, not just plausible output.
- Next: shaped real-machine equilibria (Solov'ev / Cerfon–Freidberg), q-profile, GPU scale-up on ALCF supercomputers such as Polaris and Aurora.
- Code + docs: github.com/engineer-scientist/petsc_mcp_servers_tokamak

References

- AI coding agent: *Claude Code* by Anthropic (https://claude.com/product/claude-code)
- *Automated Problem-to-Solution Generation for PDE-Based Simulation Science*, Dr Lois Curfman McInnes, Dr Junchao Zhang, Dr Matthew Knepley, et al (2026).
- *PETSc MCP Servers*, Barry Smith (2026): https://gitlab.com/petsc/petsc_mcp_servers
- *Grad-Shafronov equation* (https://en.wikipedia.org/wiki/Grad-Shafranov_equation).
- DoE Explains… Tokamaks: https://www.energy.gov/science/doe-explainstokamaks
- PETSc AI: https://petsc-ai.org
- *Improving Usability and Productivity of PETSc with Agent-Based Workflows*, Dr Barry Smith, et al (2026).

Acknowledgements

- This work was partially supported by the U.S. Department of Energy (DOE) Office of Science “Distinguished Scientist Fellows” Program.
- This work used ANL CELS General Computing Environment: https://help.cels.anl.gov/docs/linux/


Argonne
NATIONAL LABORATORY
U.S. DEPARTMENT of ENERGY